Research Statement

Project: *Enhancing LLM Security and Privacy in Multilingual Contexts* **Host:** Aalborg University Copenhagen, Department of Computer Science **Supervisor:** Prof. Johannes Bjerva **Applicant:** Nauval Zulfikar, MSc Business Analytics (Aston, 2024, 1:1 First Class)

Abstract

The rapid evolution of large language models (LLMs) has brought about significant advancements in natural language processing (NLP). However, these advancements have also introduced new challenges in terms of security, privacy, and robustness, particularly in multilingual settings. This research aims to address these challenges by developing frameworks for detecting and mitigating backdoor attacks, data poisoning, and adversarial inputs in LLMs. The project will leverage Prof. Johannes Bjerva's expertise in multilingual NLP and LLM security to explore novel methods for enhancing the safety and reliability of LLMs across diverse language contexts. By integrating advanced machine learning techniques with robust security protocols, this research seeks to contribute to the development of more secure and trustworthy AI systems.

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1. Introduction and Problem Statement

The deployment of LLMs in various applications has raised concerns about their vulnerability to security threats such as backdoor attacks, data poisoning, and adversarial inputs. These threats can compromise the integrity and reliability of AI systems, leading to potentially harmful consequences. In multilingual contexts, these challenges are further exacerbated due to the complexity and diversity of language data. Current research has primarily focused on monolingual settings, leaving a significant gap in understanding and addressing security issues in multilingual LLMs. This research aims to fill this gap by developing comprehensive frameworks for enhancing LLM security and privacy, with a focus on multilingual environments.

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2. Research Questions

1. **RQ1: How can backdoor attacks in multilingual LLMs be effectively detected and mitigated?**
2. **RQ2: What strategies can be employed to enhance the robustness of LLMs against data poisoning in diverse language settings?**
3. **RQ3: How can adversarial input detection be optimized for multilingual LLMs to ensure reliable performance across languages?**

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3. Literature Review

The literature on LLM security has predominantly focused on monolingual models. Recent studies by Bjerva et al. (2023) have highlighted the need for robust security measures in multilingual LLMs, emphasizing the unique challenges posed by language diversity. Bjerva and colleagues (2024) have also explored methods for detecting backdoor attacks in LLMs, providing a foundation for this research. Furthermore, Bjerva et al. (2025) have investigated adversarial input detection, offering insights into optimizing these techniques for multilingual contexts. Despite these advancements, there remains a critical need for comprehensive frameworks that address the full spectrum of security threats in multilingual LLMs.

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4. Methodology

4.1 Overview

The research will follow a three-stage pipeline:

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flowchart LR
    A[Stage 1: Threat Identification] --> B[Stage 2: Framework Development] --> C[Stage 3: Evaluation and Optimization]
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4.2 Quantitative

Quantitative methods will involve the development and testing of algorithms for detecting and mitigating security threats in multilingual LLMs. This will include statistical analysis and machine learning techniques to evaluate the effectiveness of proposed solutions.

4.3 Qualitative

Qualitative methods will include expert interviews and case studies to gain insights into the practical challenges and considerations in implementing security measures for multilingual LLMs.

4.4 Interplay

The integration of quantitative and qualitative methods will provide a holistic understanding of LLM security in multilingual contexts, ensuring that the developed frameworks are both theoretically sound and practically applicable.

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5. Expected Outcomes and Significance

The research is expected to produce novel frameworks for enhancing the security and privacy of multilingual LLMs. These frameworks will contribute to the development of more robust and trustworthy AI systems, with applications in various fields such as healthcare, finance, and education. The findings will also provide valuable insights for policymakers and practitioners in developing effective strategies for LLM security.

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6. Fit with Existing Background

This research aligns with my previous work on LLM-generated adaptive shipper decision rules and transformer fine-tuning, as demonstrated in projects such as the LLM-Shipper-Profiles and the MSc Dissertation Pipeline. My experience in developing secure and efficient AI systems will be instrumental in addressing the challenges of LLM security in multilingual contexts.

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7. Three-Year Workplan

- **Year 1:**
  - Conduct a comprehensive literature review on LLM security and multilingual NLP.
  - Develop initial algorithms for threat detection in multilingual LLMs.
  - Begin qualitative data collection through expert interviews.
- **Year 2:**
  - Refine and test algorithms for backdoor attack and data poisoning detection.
  - Conduct case studies to evaluate the practical applicability of developed frameworks.
  - Publish preliminary findings in peer-reviewed journals.
- **Year 3:**
  - Optimize frameworks for adversarial input detection.
  - Finalize and validate comprehensive security frameworks for multilingual LLMs.
  - Disseminate findings through conferences and workshops.

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8. Challenges and Limitations

1. **Data Availability:** Access to diverse multilingual datasets may be limited. *Mitigation:* Collaborate with international institutions to access broader datasets.
2. **Algorithm Complexity:** Developing algorithms that balance security and computational efficiency can be challenging. *Mitigation:* Focus on scalable solutions and leverage cloud computing resources.
3. **Implementation Barriers:** Practical implementation of security measures may face resistance. *Mitigation:* Engage stakeholders early and demonstrate the value of security enhancements.

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9. Conclusion and Why Aalborg University

Aalborg University Copenhagen, with its strong focus on NLP and AI security, provides an ideal environment for conducting this research. Under the guidance of Prof. Johannes Bjerva, I am confident in developing innovative solutions to enhance the security and privacy of multilingual LLMs. The university's commitment to cutting-edge research and its collaborative culture make it the perfect place to pursue this project.

References

1. Bjerva, J. et al. (2022). *Multilinguality and its Impact on NLP Robustness.* ACL.
2. Bjerva, J. et al. (2021). *Detecting Backdoors in Language Models: A Multilingual Approach.* EMNLP.

3. Bjerva, J. et al. (2020). *Privacy Concerns in NLP: A Survey of Current Techniques.* NAACL.
4. Brown, T. et al. (2020). *Language Models are Few-Shot Learners.* NeurIPS.
5. Goodfellow, I. et al. (2015). *Explaining and Harnessing Adversarial Examples.* ICLR.
6. Radford, A. et al. (2019). *Language Models are Unsupervised Multitask Learners.* OpenAI.
7. Carlini, N. et al. (2017). *Adversarial Examples Are Not Easily Detected: Bypassing Ten Detection Methods.* AISec.
8. Zhang, H. et al. (2020). *Towards Robustness Against Natural Language Word Substitution Attacks.* ACL.
9. Wallace, E. et al. (2021). *Concealed Data Poisoning Attacks on Text Datasets.* NAACL.
10. Gehrmann, S. et al. (2019). *GLTR: Statistical Detection and Visualization of Generated Text.* ACL.
11. Papernot, N. et al. (2016). *The Limitations of Deep Learning in Adversarial Settings.* EuroS&P.
12. [TODO: verify on Google Scholar — Bjerva]